Theta graphs θ_{ab3} are e-positive

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Abstract

A theta graph θ_{abc} consists of two vertices joined by three internally disjoint paths of lengths a, b and c. D. G. L. Wang conjectured that every theta graph has an e-positive chromatic symmetric function. The cases $c = 1$ (cycle-chords) and $c = 2$ (clocks) are known. This paper proves that θ_{ab3} is e-positive for every $a \geq b \geq 3$. Starting directly from Chen, He and Wang's general composition expansion, we classify its negative contributions and adapt their partial-reversal and full-fibre method to $c = 3$. A gap-vector accounting bounds the total weighted cost even when several negative terms share a target: at most one gap can have a positive deficit. We also classify exactly the zero-sum nonempty groups whose first part is at least two. Exact finite checks through 24 vertices test the stated coefficient formulas and accounting identities; the unrestricted result rests on the written proof. The case of shortest path at least four remains open here.

1 Introduction

For a finite simple graph G with vertex set V, Stanley's chromatic symmetric function [4] is $X_G = \sum_\kappa \prod_{v \in V} x_{\kappa(v)}$, the sum running over proper colourings $\kappa\colon V \to \mathbb{Z}_{>0}$. The graph G is *e-positive* if X_G is a nonnegative combination of elementary symmetric functions e_λ. The Stanley–Stembridge conjecture, now proved by Hikita [3], is the best-known instance of this question; Dahlberg, Foley and van Willigenburg [2] give background on e-positivity for broader graph classes.

For integers $a \geq b \geq c \geq 1$, the *theta graph* θ_{abc} is formed by two vertices joined by three internally disjoint paths with a, b and c edges. It has $a + b + c$ edges and $n = a + b + c - 1$ vertices. Wang [5, Conjecture 4.1] conjectured that every θ_{abc} is e-positive. Wang proved the case $c = 1$ (cycle-chords) [5], and Chen, He and Wang proved the case $c = 2$ (clocks) [1].

Theorem 1. *For all integers $a \geq b \geq 3$, the theta graph θ_{ab3} is e-positive. Hence every theta graph θ_{abc} with $c \leq 3$ is e-positive.*

The second sentence follows from the first together with [5, 1]. We use simple theta graphs, so the three paths are distinct; if the notation is extended to $b = c = 1$ by merging repeated edges, the underlying graph is a cycle (or K_2 when $a = 1$), already e-positive. The edge and vertex counts above refer to the nondegenerate theta construction.

Method and contribution. The starting point is the expansion $X_{\theta_{ab3}} = \sum_I D_I w_I e_I$ over compositions I of n, taken directly from [1, Theorem 3.1] (Section 2). The composition expansion, partial reversals and full-fibre strategy are prior work. The contribution here is their $c = 3$ implementation: an exact negative-set classification, a one-deficit gap bound and a sharp classification of zero-sum groups. The coefficients D_I can be negative. The argument has four steps.

1. The cited expansion gives an explicit prefix-sum formula and its closed forms (Lemmas 1 and 2); no removal of an involution is needed.
2. The negative set $N = \{I : D_I < 0\}$ is described exactly, and $D_I \geq -6$ (Lemma 3).
3. Each $H \in N$ is sent to a composition $\psi_3(H)$ with the same multiset of parts by reversing a prefix. The map ψ_3 is not injective on N, but the regrouping remains valid provided no target lies in N.
4. The full group sum at each target is nonnegative (Lemmas 5 and 7). The main case uses a gap vector of the target and a per-gap comparison of cost against gain.

Section 8 lists the exact computations, and Section 9 explains the boundary of the result.

2 Preliminaries

2.1 Compositions

A composition $I = i_1 i_2 \cdots i_z$ of n is a sequence of positive integers with sum $|I| = n$; $\ell(I) = z$. Write $\overline{I} = i_z \cdots i_1$ for the reversal, $e_I = e_{i_1} \cdots e_{i_z}$, and $P_I(k) = i_1 + \cdots + i_k$ for prefix sums, with $P_I(0) = 0$. Following [1], put

$$w_I = i_1(i_2 - 1)(i_3 - 1) \cdots (i_z - 1).$$

Only compositions with $w_I \neq 0$ contribute below; for these, $i_j \geq 2$ for all $j \geq 2$. For integers $x \in [0, n]$,

$$\Theta_I^+(x) = \min\{P_I(k) : 0 \leq k \leq z,\ P_I(k) \geq x\} - x,$$
$$\Theta_I^-(x) = x - \max\{P_I(k) : 0 \leq k \leq z,\ P_I(k) \leq x\},$$

and $\Theta_I^-(x) = \Theta_{\overline{I}}^+(n - x)$ [1, Lemma 2.5]. Let $e_2(y_1, \ldots, y_k) = \sum_{i<j} y_i y_j$.

Definition 1 ([5, 1])**.** For $1 \leq b' < n$, write $b' = i_1 + \cdots + i_{p-1} + s = i_2 + \cdots + i_q + t$ with $1 \leq p, q \leq z$, $1 \leq s \leq i_p$, $1 \leq t \leq i_{q+1}$, and $i_{z+1} = i_1$. Define

$$\Delta_I(b') = \begin{cases} s(i_p - s - i_1), & i_1 \leq i_p - s, \\ e_2(i_p - s,\, i_{p+1}, \ldots, i_q,\, t), & \text{otherwise.} \end{cases}$$

2.2 The input expansion

Fix $a \geq b \geq 3$, put $n = a + b + 2$ and $B = b + 2$. Then $B \geq 5$ and $n - B = a$. The following identity is [1, Theorem 3.1] specialised to $c = 3$, using reflection of prefix sums:

$$X_{\theta_{ab3}} = \sum_{I \models n} D_I w_I e_I, \qquad D_I = \Theta_I^+(2) + \Theta_I^+(3) - \Theta_I^+(B) - \Theta_I^+(B-1) + \Delta_I(B). \tag{1}$$

This is the sole input taken from the literature. Identity (1) was also compared with a brute-force computation of $X_{\theta_{ab3}}$ for nine pairs (a, b) (Section 8), with exact agreement.

2.3 Notation for a fixed composition

For I with $w_I \neq 0$ write $x = i_1$. Let p be the index of the part containing B, that is $P_I(p-1) < B \leq P_I(p)$, and put $\sigma = P_I(p-1)$, $s = B - \sigma$ and $r = i_p - s \geq 0$, so $1 \leq s \leq i_p$ and $P_I(p) = B + r$. These p, s coincide with those in the definition of $\Delta_I(B)$.

3 Reduction to prefix sums

Lemma 1. *For every I with $w_I \neq 0$,*

$$D_I = \Theta_I^+(2) + \Theta_I^+(3) - \Theta_I^+(B-1) - \Theta_I^+(B) + \Delta_I(B).$$

Proof. This is exactly (1). In Chen, He and Wang's notation the negative sum is $\Theta_{\overline{I}}^-(a) + \Theta_{\overline{I}}^-(a+1)$; the reflection identity gives $\Theta_I^+(B) + \Theta_I^+(B-1)$. Their alternative involution-based Theorem 3.2 gives the same coefficients when $c = 3$, but that additional fact is not needed here. □

Lemma 2. *Let $T(I) = \Theta_I^+(2) + \Theta_I^+(3)$ and $U(I) = \Theta_I^+(B-1) + \Theta_I^+(B)$. Then*

$$T = \begin{cases} 2i_2 - 3, & x = 1, \\ i_2 - 1, & x = 2, \\ 2x - 5, & x \geq 3, \end{cases} \qquad U = \begin{cases} 2r + 1, & s \geq 2, \\ r, & s = 1. \end{cases}$$

Consequently:

1. *If $p = 1$ (that is, $B \leq x$), then $D_I = 2B - 6 + \Delta_I(B) \geq 4$.*
2. *If $p \geq 2$ and $r \geq x$ (branch 1), then $\Delta_I(B) = s(r-x)$, and $D_I = T - x$ if $s = 1$, while $D_I = T - 2x - 1 + (s-2)(r-x)$ if $s \geq 2$.*
3. *If $p \geq 2$ and $r < x$ (branch 2), then $\Delta_I(B) = e_2(r, i_{p+1}, \ldots, i_q, t)$, the arguments form a weak composition of x, and $D_I = T - U + \Delta_I(B)$.*

Proof. T is read off from the first one or two prefix sums; for $x = 1$ one uses $1 + i_2 \geq 3$. The least prefix sum at least B is $P(p) = B + r$; the least prefix sum at least $B - 1$ is σ when $s = 1$ and $P(p)$ otherwise, which gives U. For $p = 1$, $T = 2x - 5$ and $U = 2(x - B) + 1$ (as $s = B \geq 2$), so $D = 2B - 6 + \Delta \geq 4$ by $B \geq 5$ and $\Delta \geq 0$. Branch 1 is the first case of Δ, and $r \geq x$ forces $p \geq 2$. In branch 2, from $B = i_2 + \cdots + i_q + t$ one gets $B + x = P(q) + t$, so $P(q) < B + x \leq P(q+1)$ (indices cyclic). Since $B + x > P(p)$, $q \geq p$, and $r + (i_{p+1} + \cdots + i_q) + t = (P(p) - B) + (P(q) - P(p)) + (B + x - P(q)) = x$. □

4 The negative set

Lemma 3. *Let $w_I \neq 0$. Then $D_I < 0$ exactly in the following cases, all with $p \geq 2$:*

N1 *$s = 1$, $r \geq x$, and*

N2 *$s \geq 2$, $r = x - 1$; in both, $D_I = T - x$, which is negative exactly when $x = 2$, $i_2 = 2$ ($D = -1$), or $x = 3$ ($D = -2$), or $x = 4$ ($D = -1$);*

N3 *$s \geq 2$, $r \geq x$, with $D_I = T - 2x - 1 + (s-2)(r-x) < 0$; explicitly: for $x = 1$, $i_2 = 2$ and $(s-2)(r-1) \leq 1$, so $D \in \{-2, -1\}$; for $x = 2$, $i_2 - 6 + (s-2)(r-2) < 0$; for $x \geq 3$, $(s-2)(r-x) \leq 5$ and $D = -6 + (s-2)(r-x)$.*

In particular $D_I \geq -6$ for all I.

Proof. N1 and N3 follow from Lemma 2(2). For N1 with $x = 1$, $D = 2i_2 - 4 \geq 0$, so N1 requires $x \geq 2$; for $x \geq 5$, $T - x = x - 5 \geq 0$. For N2, branch 2 with $r = x - 1$ forces $q = p$ and $t = 1$, so $\Delta = x - 1$ and $D = T - 2(x-1) - 1 + (x-1) = T - x$. The remaining cases are nonnegative:

- $s \geq 2$, $r = 0$: for $x \in \{1, 2\}$ the weak composition after r is just $t = x$, so $\Delta = 0$ and $D = T - 1$, which is $2i_2 - 4 \geq 0$ or $i_2 - 2 \geq 0$; for $x \geq 3$, $D = 2x - 6 + \Delta \geq 0$.
- $s \geq 2$, $1 \leq r \leq x - 2$ (so $x \geq 3$): $\Delta \geq r(x - r)$, so $D \geq f(r) := 2x - 6 - 2r + r(x - r)$; f is concave with $f(1) = 3x - 9 \geq 0$ and $f(x - 2) = 2x - 6 \geq 0$.
- $s = 1$, $r < x$: then $r = i_p - 1 \geq 1$, so $x \geq 2$. For $x = 2$, $r = 1$, $\Delta = 1$ and $D = i_2 - 1 \geq 1$. For $x \geq 3$, $D \geq 2x - 5 + r(x - r - 1) > 0$.
- $p = 1$: $D \geq 4$ by Lemma 2(1). □

5 Regrouping

For a composition H and threshold y, let $p = p(H; y)$ be the index of the part containing y and set $\psi(H; y) = \overline{(h_1 \cdots h_p)}\, h_{p+1} \cdots h_z$. Define

$$\psi_3(H) = \begin{cases} (1) \cdot \psi(h_2 h_3 \cdots ;\, B - 1), & h_1 = 1, \\ \psi(H; B), & h_1 \geq 2. \end{cases}$$

Since $\psi_3(H)$ is a rearrangement of H, $e_{\psi_3(H)} = e_H$. Hence

$$X_{\theta_{ab3}} = \sum_{J \notin N} c_J e_J, \qquad c_J = D_J w_J + \sum_{H \in N,\ \psi_3(H) = J} D_H w_H, \tag{2}$$

provided $\psi_3(N) \cap N = \emptyset$: each nonnegative summand stays in place and each negative summand is moved to an identical symmetric-function term indexed by its target. Injectivity is not required, and collecting composition-level coefficients by partitions preserves nonnegativity. Theorem 1 therefore reduces to showing that no target lies in N and that every $c_J \geq 0$.

Remark 1. The map ψ_3 is not injective on N. For $(a, b) = (4, 3)$, both $(2, 2, 5)$ and $(2, 5, 2)$ map to $(5, 2, 2)$. The target has $D_J w_J = 12 \cdot 5 = 60$, while each source contributes $-1 \cdot 8 = -8$, leaving $c_J = 44$. For $(a, b) = (8, 7)$, the compositions $(2, 2, 2, 2, 7, 2)$, $(2, 2, 2, 7, 2, 2)$ and $(2, 2, 7, 2, 2, 2)$ all map to $(7, 2, 2, 2, 2, 2)$. It is the entire weighted group, not any one pair, that must be nonnegative.

Lemma 4 (Preimage structure)**.** *Fix a target J with $J_1 = m$. For $j \geq 2$ let $H_j = \overline{(J_1 \cdots J_j)}\, J_{j+1} \cdots$, so that $\sigma(H_j) = P_J(j) - m$, $s(H_j) = B + m - P_J(j)$ and $r(H_j) = P_J(j) - B$ whenever $p(H_j) = j$. Suppose $h_1 \geq 2$.*

1. *$p(H_j) = j$ if and only if the window condition $B \leq P_J(j) \leq B + m - 1$ holds. This condition implies $\psi(H_j; B) = J$. Every negative preimage is obtained at its actual index $j = p(H)$; distinct admissible indices give distinct preimages. The converse from $\psi(H_j; B) = J$ alone need not hold, because a longer reversed prefix may be palindromic.*
2. *Among such H_j, the N2 branch ($r_H = x_H - 1$ with $s_H \geq 2$) occurs exactly when $j = p(J)$, $P_J(j - 1) = B - 1$ and $m \geq J_j + 1$ (equivalently $s_H = m - J_j + 1 \geq 2$).*
3. *Among such H_j, the N1/N3 branch ($r_H \geq x_H$) occurs exactly when $P_J(j - 1) \geq B$.*
4. *$w_{H_j}/w_J = J_j(m - 1)/(m(J_j - 1)) < 2$, and $w_J \neq 0$.*

Membership $H_j \in N$ additionally requires $D_{H_j} < 0$, which is decided by Lemma 3; the branch conditions alone do not imply negativity. For example, with $B = 10$ and $J = (11, 9)$, $H_2 = (9, 11)$ satisfies the conditions in (3) but $D_{H_2} = 4 > 0$.

Proof. (1) The block $\overline{(J_1 \cdots J_j)}$ ends with $J_1 = m$ at position j of H_j, and $P_{H_j}(j) = P_J(j)$, $P_{H_j}(j-1) = P_J(j) - m$. Thus $p(H_j) = j$ iff $P_J(j) - m < B \le P_J(j)$; reversing this actual prefix then gives J. Every negative H with $\psi(H; B) = J$ has this form with $j = p(H)$, and $j \ge 2$ since $p(H) = 1$ forces $D_H \ge 4$. Two admissible indices cannot give the same composition, since its actual index $p(H)$ is unique. (2)–(3) In H_j, $x_H = J_j$ and $r_H = P_J(j) - B$, so $r_H - x_H = P_J(j-1) - B$. Hence $r_H \ge x_H$ iff $P_J(j-1) \ge B$, and $r_H = x_H - 1$ iff $P_J(j-1) = B - 1$, which says $j = p(J)$ with $s(J) = 1$. The N2 branch also needs $s_H = m - r_H = m - J_j + 1 \ge 2$. (4) H_j and J have the same parts; the first part of J is m and that of H_j is J_j, so the ratio is $J_j(m-1)/(m(J_j-1))$. All parts of H are at least 2, including h_1 by hypothesis. The same is true of its rearrangement J, so $w_J \ne 0$ and $m \ge 2$. The ratio is at most $J_j/(J_j - 1) \le 2$ times $(m-1)/m < 1$. □

Lemma 5 (Groups with $h_1 = 1$)**.** *Let J be the image under ψ_3 of some $H \in N$ with $h_1 = 1$. Then $D_J \ge 0$ (so $J \notin N$) and $c_J \ge 0$.*

Proof. Write $H = (1, K)$ and measure positions in K relative to $B' = B - 1$, with part index p' and parameters s, r. By Lemma 2 (with $x = 1$), $D_H = 2k_1 - 4$ if $s = 1$ or $r = 0$, and $D_H = 2k_1 - 6 + (s-2)(r-1)$ otherwise. Hence $H \in N$ exactly when $p' \ge 2$, $k_1 = 2$, $s \ge 2$, $r \ge 1$ and $(s-2)(r-1) \le 1$; then $D_H \in \{-2, -1\}$. The target is $J = (1, K')$ with $K' = \overline{(k_1 \cdots k_{p'})} \cdot k_{p'+1} \cdots$, and $w_J = w_H$ because parts of index at least 2 are only permuted.

Put $m = K'_1 = s + r \ge 3$. Applying Lemma 4(1) to K' with threshold B', every preimage corresponds to an index j with $K'_j = 2$ (as $k_1 = 2$) and $r_j = P_{K'}(j) - B' \in [1, m-2]$. Distinct preimages have prefix sums $P_{K'}(j)$ differing by at least 2 (all parts after the first are at least 2), so their r-values differ by at least 2. If $r_j \ge 3$, then $(s_j - 2)(r_j - 1) \le 1$ forces $s_j = 2$, so $r_j = m - 2$. The possible sets of r-values are therefore $\{1\}$, $\{2\}$, $\{m-2\}$, $\{1, m-2\}$ with $m \ge 5$, and $\{2, m-2\}$ with $m \ge 6$. The last is impossible: the member with $r = 2$ needs $(s-2)(r-1) = m - 4 \le 1$, i.e. $m \le 5$.

Let j_0 be the smallest preimage index. D_J is computed by Lemma 2 with $x = 1$, $i_2 = m$:

- $r_{j_0} = 1$: $P_{K'}(j_0 - 1) = B' - 1$, so $s_J = 1$ and $D_J = 2m - 4 \ge 2$; with two preimages $m \ge 5$ and $D_J \ge 6$.
- $r_{j_0} = 2$: B' is a prefix sum of K', so $r_J = 0$ and $D_J = 2m - 4 \ge 4$ (here $m \ge 4$).
- $r_{j_0} \ge 3$: $D_J \ge 2m - 6 \ge 4$.

Every $|D_H| \le 2$ and all weights in the group are equal, so $c_J/w_J = D_J + \sum D_H \ge 0$ in each case, and $D_J \ge 0$. □

6 Groups with first part at least two

Lemma 6 (Separated suffixes and one deficit)**.** *Let $g_1, \ldots, g_L \ge 2$ and $t \ge 1$ be integers, and set $S_i = g_{i+1} + \cdots + g_L + t$. If nonnegative costs u_i satisfy $u_i \le g_i S_i$ except possibly when $S_i \in \{2, 3\}$, then at most one index can have a positive deficit. If its deficit is at most $d \ge 0$, then $\sum_i u_i \le \sum_i g_i S_i + d$.*

Proof. Successive suffixes differ by $g_{i+1} \ge 2$, so two suffixes cannot both belong to $\{2, 3\}$. Sum the nonpositive deficits and the possible single positive one. The hypotheses on costs will be verified below, rather than assumed for the graph coefficients. □

Lemma 7. *Let J be the target of some $H \in N$ with $h_1 \ge 2$. Then*

(i) $D_J > 0$; in particular $J \notin N$;

(ii) $c_J \geq 0$.

Remark 2. For $(a, b) = (4, 4)$ and $J = (4, 2, 2, 2)$ the source $(2, 2, 4, 2)$ gives exact cancellation: $6 \cdot 4 - 4 \cdot 6 = 0$. Proposition 1 below classifies all such zero-sum nonempty groups with first part at least two. This is a statement about the regrouping, not a classification of zero partition coefficients.

Proof of Lemma 7. Setup. Put $m = J_1$. By Lemma 4, each H in the group equals H_j for some $j \geq 2$ satisfying $B \leq P_J(j) \leq B + m - 1$. For H_j: $x = y := J_j$, $i_p = m$, $i_2(H_j) = J_{j-1}$, $r_H = P_J(j) - B$, $s_H = m - r_H$, and $w_{H_j} = \rho(y) w_J$ with $\rho(y) = y(m-1)/(m(y-1))$. All J_i with $i \geq 2$ are at least 2.

Let $k = p(J)$, $s_J = s(J)$ and $r = r(J) = P_J(k) - B$. If $r \geq m$, no j satisfies the window condition with $H_j \in N$: $j = k$ gives $s_H \leq 0$ and $j > k$ gives $P_J(j) \geq B + m$. Hence $r \leq m - 1$, and J lies in branch 2 of Lemma 2 (also when $k = 1$, since then $r = m - B < m$). Let $q \leq \ell(J)$ be maximal with $P_J(q) < B + m$ and $t = B + m - P_J(q) \geq 1$. The *gap vector* is

$$g = (r, g_1, \ldots, g_L, t), \qquad g_i = J_{k+i} \geq 2, \quad L = q - k, \qquad r + \textstyle\sum g_i + t = m,$$

and $\Delta_J(B) = e_2(g)$. If $B + m > n$, the terminal amount t lies in the cyclic copy of the first part; no full part beyond index $\ell(J)$ is introduced. Put $R_i = r + g_1 + \cdots + g_i$ and $S_i = g_{i+1} + \cdots + g_L + t$, so $S_0 = m - r$ and $S_L = t$.

Classification of preimages. By Lemma 3, $H_j \in N$ requires $r_H \geq y$ (N1/N3) or $r_H = y - 1$ with $s_H \geq 2$ (N2). By Lemma 4, $r_H \geq y$ iff $j \geq k + 1$, and the window gives $j \leq q$; and $r_H = y - 1$ iff $j = k$ and $s_J = 1$ (so $k \geq 2$). The candidates are:

- *N2 at $j = k$*: needs $k \geq 2$, $s_J = 1$, $S_0 \geq 2$, and $y = J_k = r + 1$. By Lemma 3, $D_H < 0$ only for $y = 2$ with $J_{k-1} = 2$ ($D = -1$), $y = 3$ ($D = -2$), $y = 4$ ($D = -1$). Its cost $\rho(y)|D_H|$ satisfies $\mathrm{cost}_{N2} \leq 3(m-1)/m < 3$.

- *Gap index $i \in \{1, \ldots, L\}$ ($j = k + i$)*: $y = g_i$, $r_H = R_i$, $s_H = S_i$, $r_H - y = R_{i-1}$. The type is N1 iff $i = L$ and $t = 1$; otherwise N3, with $D_H = -6 + (S_i - 2)R_{i-1}$ for $y \geq 3$ and $D_H = J_{j-1} - 6 + (S_i - 2)R_{i-1}$ for $y = 2$; in the latter case $J_{j-1} \geq 2$ gives $-D_H \leq 4$.

Gap decomposition of D_J. If $m = 2$ then $L = 0$ and $r + t = 2$; N2 would need $r \geq 1$ and $S_0 = 2 - r \geq 2$, which is impossible, so there are no preimages. Let $m \geq 3$; then $T(J) = 2m - 5$, and $U(J) = r$ if $s_J = 1$ and $2r + 1$ otherwise (also for $k = 1$, where $s_J = B \geq 2$). Expanding $e_2(g)$ as each entry times the sum of later entries,

$$D_J = A + \sum_{i=1}^{L} g_i S_i, \qquad A := 2m - 5 - U + r(m - r),$$

so $A = 2m - 6 + r(m - r - 2)$ if $s_J \geq 2$ and $A = 2m - 5 + r(m - r - 1)$ if $s_J = 1$.

Per-gap bound. For a gap index i let $\mathrm{cost}_i = \rho(y) \max(0, -D_{H_j})$ and $\mathrm{def}_i = \mathrm{cost}_i - g_i S_i$. Use $\rho(y) \leq y/(y-1)$ and $(m-1)/m < 1$.

- N1 ($S_L = 1$): for $y = 2$, cost $\leq 2(m-1)/m < 2 = g_i S_i$; for $y = 3$, cost $= 3(m-1)/m < 3$; for $y = 4$, cost $< 4/3 < 4$; for $y \geq 5$, $D_H = y - 5 \geq 0$. No deficit.

- N3, $y \geq 3$ (so cost $\leq 6\rho(y)$): if $S_i = 2$, $\mathrm{def}_i \leq 6y(m-1)/(m(y-1)) - 2y$, which equals $3 - 9/m$ for $y = 3$ and is negative for $y \geq 4$; if $S_i = 3$, $\mathrm{def}_i \leq 6y/(y-1) - 3y = -3y(y-3)/(y-1) \leq 0$; if $S_i \geq 4$, $g_i S_i \geq 4y \geq 6y/(y-1) \geq \mathrm{cost}_i$.

- N3, $y = 2$ (cost $\leq 4\rho(2) = 8(m-1)/m$): $S_i = 2$ gives $\mathrm{def}_i \leq 4 - 8/m$; $S_i = 3$ gives $\mathrm{def}_i \leq 2 - 8/m$; $S_i \geq 4$ gives $g_i S_i \geq 8 > \mathrm{cost}_i$.

Thus a positive deficit requires an N3 index with $S_i \in \{2,3\}$, the setting of Lemma 6. Since $S_L = t$ and $S_{L-1} = g_L + t \geq t+2$, this happens for at most one index: $i = L$ if $t \in \{2,3\}$, or $i = L-1$ if $t = 1$ and $g_L = 2$ (the index L then being N1). Let δ be that deficit ($\delta = 0$ if there is none). Then $\delta \leq 4 - 8/m < 4$, and the deficit index has $y + S_i \leq m - r$.

Conclusion. Every non-deficit gap index, whether or not it carries a preimage, has $g_i S_i \geq \mathrm{cost}_i \geq 0$, so

$$c_J/w_J = D_J - \mathrm{cost}_{N2} - \sum_i \mathrm{cost}_i \;\geq\; A - \mathrm{cost}_{N2} - \delta.$$

If $\delta > 0$ or a gap preimage exists, then $L \geq 1$, so $m - r \geq 2 + 1 = 3$ and $r(m-r-2) \geq 0$.

- $s_J \geq 2$ (this includes $k = 1$; there is no N2 preimage): $A \geq 2m - 6$. A deficit with $y = 2$, $S_i = 2$ needs $m \geq 4$, and $2m - 6 - (4 - 8/m) = 2(m-1)(m-4)/m \geq 0$. The deficits with $y = 3, S_i = 2$ or $y = 2, S_i = 3$ need $m \geq 5$, where $A \geq 4 > 3 > \delta$. Without a deficit, $A \geq 0$ when $L \geq 1$, and when $L = 0$ there are no gap preimages and no N2 preimage, so the group is empty.

- $s_J = 1$ without N2: $A \geq 2m - 5$, which is at least $4 - 8/m$ for $m \geq 4$, and at least 3 for $m \geq 5$ when $y + S_i \geq 5$; with no deficit $A \geq 1$.

- $s_J = 1$ with N2 ($r = y - 1 \in \{1,2,3\}$): if $L \geq 1$, then $m - r \geq 3$, so $r(m-r-1) \geq 2r$ and $A \geq 2m - 3$. With a deficit, $m \geq r + g_i + S_i \geq 5$, so $A \geq 7$; since $\mathrm{cost}_{N2} < 3$ and $\delta \leq 4 - 8/m < 4$ strictly, $\mathrm{cost}_{N2} + \delta < 7 \leq A$. Without a deficit, $A \geq 5 > 3 > \mathrm{cost}_{N2}$. If $L = 0$ (then $S_0 = t \geq 2$ and there are no gap preimages), $A = 2m - 5 + r(m-r-1)$ equals $3m - 7 \geq 2 > 2(m-1)/m$ for $r = 1$, equals $4m - 11 \geq 5 > 3$ for $r = 2$, and is at least 8 for $r = 3$; in each case $A > \mathrm{cost}_{N2}$.

Hence $c_J \geq 0$, proving (ii). For (i): the group is nonempty and every $D_H w_H < 0$, so $D_J w_J \geq \sum_H |D_H| w_H > 0$. □

Proposition 1 (Sharp zero-group classification)**.** *For a nonempty group in Lemma 7, $c_J = 0$ if and only if*

$$m = 4, \quad r = 0, \quad L = 1, \quad g = (0,2,2), \quad J_k = 2.$$

Proof. Every $s_J = 1$ case in the proof is strict. Without N2, $2m - 5$ strictly exceeds each possible positive deficit; with N2 the displayed estimates are strict. If $s_J \geq 2$, deficit types $(3,2)$ and $(2,3)$ require $m \geq 5$ and are strictly below $A \geq 4$. Type $(2,2)$ leaves the lower bound $2(m-1)(m-4)/m$, positive unless $m = 4$. At $m = 4$, the total gap length forces $r = 0$, $L = 1$, $g_1 = t = 2$. Here $A = 2$, the sole gain is 4, and its candidate source has coefficient $J_k - 6$ and weight ratio $3/2$. Consequently

$$c_J/w_J = 6 - \tfrac{3}{2}\max(0, 6 - J_k),$$

which is zero exactly when $J_k = 2$, since $J_k \geq 2$. Conversely these values give a negative source of coefficient -4 and exact cancellation. With no positive deficit, $A > 0$ for $m \geq 4$. The only remaining nonempty possibility $m = 3$ forces $(r, g_1, t) = (0, 2, 1)$; its N1 cost is at most $4/3$, strictly less than its gain 2. Thus no further equality is possible. □

7 Proof of Theorem 1

By (1), $X_{\theta_{ab3}} = \sum_I D_I w_I e_I$. By Lemma 5 and Lemma 7(i), no target of ψ_3 lies in N, so the regrouping (2) is valid. Every $J \notin N$ that is not a target has $c_J = D_J w_J \geq 0$, and every target has $c_J \geq 0$ by Lemma 5 and Lemma 7(ii). Collecting by partitions, the coefficient of each e_λ in $X_{\theta_{ab3}}$ is a sum of nonnegative numbers. □

8 Computational verification

All computations use exact integer or rational arithmetic. The scripts are `code/verify_all.py`, `code/lemma6_check.py` and `code/verify_csf.py`, with shared definitions in `code/lib3.py`.

- *Input identity.* `verify_csf.py` computes $X_{\theta_{ab3}}$ by brute force (subset expansion and conversion to the e-basis) and compares it with (1) for the nine pairs $(a,b) = (3,3)$, $(4,3)$, $(5,3)$, $(4,4)$, $(6,3)$, $(5,4)$, $(7,3)$, $(6,4)$, $(5,5)$, i.e. $n \le 12$. All agree.

- *Lemmas 1–5 and final nonnegativity.* `verify_all.py 24` covers all 81 pairs $3 \le b \le a$ with $n \le 24$: 971,165 compositions with $w_I \neq 0$ and 170,224 groups (45,899 with $h_1 = 1$, 124,325 with $h_1 \ge 2$). It checks Lemmas 1, 2 and 3 on every composition, the preimage bounds of Lemma 5, and $c_J \ge 0$ for every group. No failures.

- *Lemma 7.* `lemma6_check.py 24` checks, for all 124,325 groups with $h_1 \ge 2$, that the predicted preimage set equals the actual one, the weight ratios, the decomposition $D_J = A + \sum g_i S_i$, each per-gap and N2 bound, that at most one deficit occurs, $A - \mathrm{cost}_{N2} - \delta \ge 0$, $D_J > 0$ and $c_J \ge 0$. It finds 65 distinct preimage patterns, all covered by the proof (which does not enumerate them), and 3,926 groups with $c_J = 0$, all of the shape in Proposition 1. No failures.

Independently of the lemmas, the final check shows that θ_{ab3} is e-positive for all $n \le 24$.

9 Scope and larger shortest paths

The general expansion is already available in Chen, He and Wang's Theorem 3.1. With $B = b + c - 1$ it reads

$$X_{\theta_{abc}} = \sum_{I \models a+b+c-1} \widehat{D}_I w_I e_I, \qquad \widehat{D}_I = \Delta_I(B) + \sum_{k=2}^{c} \Theta_I^+(k) - \sum_{k=B-c+2}^{B} \Theta_I^+(k).$$

Their proof also supplies the equivalent expansion

$$C_I = \Delta_I(b) + \sum_{k=2}^{c} \Theta_I^+(k) - \sum_{k=a+1}^{a+c-1} \Theta_I^+(k).$$

Equality of the resulting symmetric functions does not require pointwise equality of composition coefficients: rearrangements index the same e_λ. Thus failure of pointwise agreement with their involution-based formula for $c \ge 4$ is not a failure of the general expansion.

Writing $\sigma_I^+(x) = x + \Theta_I^+(x)$ shows that $C_I < 0$ requires at least one k with $\sigma_I^+(k+a-1) - \sigma_I^+(k) > a - 1$, since $\Delta_I(b) \ge 0$. This jump may traverse several parts; it need not identify one long part. For θ_{422} and $I = (2,2,3)$ one has $C_I = -2$ although every part is smaller than $a = 4$.

The present proof does not establish positivity for $c \ge 4$. Earlier exploratory suffix-dominance and max-flow claims are not used or retained as evidence: their generating scripts and complete receipts were absent from the submitted archive. A larger-c proof would require adequate control of full weighted preimages, not a new starting expansion. The one-deficit mechanism proved here is specific to the stated coefficient bounds; its usefulness for a second graph family has not been established.

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